

MADISON, MN, USA

WMO: 727466

Lat: 44.986N

Lon: 96.178W

Elev: 1082

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-8.3	-19.7	1.9	-11.1	-15.5	2.5	-7.1	32.3	14.7	29.1	17.6	6.8	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	90.8	74.0	88.3	73.2	84.5	71.3	78.6	86.2	76.5	84.5	74.4	81.9	12.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	138.9	82.6	73.3	128.9	80.9	72.2	124.1	79.5	43.0	88.1	40.6	84.6	38.7	81.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	25.1	22.2	-19.3	96.4	6.4	3.5	-24.0	98.9	-27.7	101.0	-31.3	102.9	-36.0	105.4		
(5)			-19.8	81.7	6.3	3.4	-24.3	84.1	-28.0	86.1	-31.5	88.0	-36.0	90.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	15.7	17.0	30.6	45.9	58.3	68.4	72.6	69.2	61.8	47.9	34.5	19.8
	DBStd	22.71	13.50	12.66	12.70	10.30	8.90	6.78	5.96	5.97	8.62	9.51	11.25	12.26
	HDD50	4391	1064	925	609	194	26	0	0	0	9	154	475	935
	HDD65	7848	1529	1345	1066	578	247	40	10	27	159	533	914	1400
	CDD50	2672	0	0	7	69	282	553	701	594	364	91	11	0
	CDD65	656	0	0	0	3	38	143	246	156	65	5	0	0
	CDH74	6745	0	0	1	76	507	1472	2464	1405	732	86	2	0
	CDH80	2291	0	0	0	19	183	510	900	424	235	20	0	0
(14) Wind	WSAvg	9.5	10.1	10.6	10.6	11.5	11.2	9.0	6.6	6.5	8.0	9.5	10.2	10.2
(15) Precipitation	PrecAvg	26.1	0.7	0.7	1.4	2.6	3.0	4.1	3.4	3.0	2.9	2.5	1.2	0.7
	PrecMax	33.5	2.4	1.7	3.3	6.9	4.8	8.8	6.3	6.3	8.1	7.9	3.5	2.0
	PrecMin	18.9	0.0	0.1	0.4	0.7	0.6	1.3	0.8	0.6	0.6	0.2	0.0	0.0
	PrecStd	3.9	0.6	0.4	0.7	1.4	1.3	2.0	1.5	1.3	1.8	2.0	1.1	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	52.1	70.2	81.9	91.3	93.2	94.9	91.4	91.0	82.4	70.3	51.6
		MCWB	39.9	43.7	53.7	61.9	66.4	72.0	79.3	76.4	70.9	60.2	54.4	42.1
	2%	DB	40.8	44.8	61.1	74.7	84.4	90.0	90.7	88.3	86.4	75.0	63.2	44.7
		MCWB	36.4	38.9	51.1	56.3	64.9	71.0	77.4	75.3	69.7	58.9	51.4	38.3
	5%	DB	37.1	38.9	54.1	69.7	79.5	85.7	88.2	84.1	81.8	69.9	56.6	38.9
		MCWB	34.4	35.2	47.3	53.8	62.6	70.9	76.0	73.0	66.8	56.6	46.5	35.0
	10%	DB	35.5	35.9	47.6	63.6	74.6	81.9	84.3	81.5	78.7	63.8	51.6	36.3
		MCWB	33.4	33.5	43.3	51.0	60.1	68.0	73.2	70.8	65.1	53.2	44.4	33.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	43.9	57.3	63.4	71.9	78.2	83.7	80.9	75.5	66.5	58.1	43.8
		MCDB	46.5	49.5	64.7	76.7	83.4	87.4	90.2	86.7	84.3	75.5	67.1	48.3
	2%	WB	37.2	38.8	52.1	58.9	68.3	75.5	80.1	77.3	73.2	61.6	51.8	38.3
		MCDB	40.1	43.5	59.9	71.0	79.3	84.5	86.7	85.6	81.4	71.4	60.8	42.4
	5%	WB	34.9	35.6	47.2	55.6	65.0	73.1	77.6	75.0	70.7	57.9	47.8	36.0
		MCDB	37.4	38.9	53.0	66.2	75.8	82.1	85.9	82.3	78.5	66.4	55.6	38.8
	10%	WB	32.6	32.8	42.6	52.8	62.4	70.7	75.4	72.7	67.4	54.6	44.0	33.8
		MCDB	34.8	35.3	47.9	61.5	71.7	79.2	82.3	79.1	75.3	62.5	50.3	36.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	17.9	17.1	20.6	21.3	20.2	20.7	21.2	23.5	20.9	19.0	16.1
		MCDBR	20.1	21.3	26.6	31.9	28.8	25.4	22.7	24.2	28.1	29.9	28.6	20.7
	5% WB	MCWBR	16.7	16.9	17.8	16.8	14.0	12.3	12.5	13.3	13.7	15.7	17.9	16.1
		MCDWR	18.5	20.5	25.2	26.4	24.6	20.9	20.8	21.6	22.6	24.2	26.5	18.3
(40) Clear-Sky Solar Irradiance	taub		0.267	0.278	0.311	0.359	0.378	0.389	0.395	0.400	0.360	0.319	0.284	0.266
		taud	2.441	2.421	2.429	2.335	2.351	2.378	2.349	2.314	2.410	2.504	2.517	2.454
	Ebn at Noon		274	291	294	286	283	279	275	270	273	272	266	262
		Edh at Noon	22	27	31	38	38	37	38	38	32	25	21	20
(44) All-Sky Solar Radiation	RadAvg		571	909	1204	1515	1746	1948	2086	1731	1343	869	570	421
	RadStd		64	101	117	135	128	127	77	145	98	114	55	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page